2840. Check if Strings Can be Made Equal With Operations II

Solved

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You are given two strings `s1` and `s2`, both of length `n`, consisting of **lowercase** English letters.

You can apply the following operation on **any** of the two strings **any** number of times:

- Choose any two indices `i` and `j` such that `i < j` and the difference `j - i` is **even**, then **swap** the two characters at those indices in the string.

Return `true` *if you can make the strings* `s1` *and* `s2` *equal, and* `false` *otherwise.*

Example 1:

```
Input: s1 = "abcdba", s2 = "cabdab"
Output: true
Explanation: We can apply the following operations on s1:
- Choose the indices i = 0, j = 2. The resulting string is s1 = "cbadba".
- Choose the indices i = 2, j = 4. The resulting string is s1 = "cbbdaa".
- Choose the indices i = 1, j = 5. The resulting string is s1 = "cabdab" = s2.
```

Example 2:

```
Input: s1 = "abe", s2 = "bea"
Output: false
Explanation: It is not possible to make the two strings equal.
```

Constraints:

- `n == s1.length == s2.length`
- $1 <= n <= 10^5$
- `s1` and `s2` consist only of lowercase English letters.

Python:

```
class Solution:
```

```python
    def checkStrings(self, s1: str, s2: str) -> bool:
        freq = [0] * 52

        for i, (a, b) in enumerate(zip(s1, s2)):
            off = (i & 1) * 26
            freq[ord(a) - 97 + off] += 1
            freq[ord(b) - 97 + off] -= 1

        return all(c == 0 for c in freq)
```

JavaScript:

```javascript
const checkStrings = (s1, s2) => {
    const freq = Array(52).fill(0);

    for (let i = 0; i < s1.length; i++) {
        const off = (i & 1) * 26;
        freq[s1.charCodeAt(i) - 97 + off]++;
        freq[s2.charCodeAt(i) - 97 + off]--;
    }

    return freq.every(c => c === 0);
};
```

Java:

```java
class Solution {
    public boolean checkStrings(String s1, String s2) {
        int[] freq = new int[52];

        for (int i = 0; i < s1.length(); i++) {
            int off = (i & 1) * 26;
            freq[s1.charAt(i) - 'a' + off]++;
            freq[s2.charAt(i) - 'a' + off]--;
        }

        for (int i = 0; i < 52; i++)
            if (freq[i] != 0)
                return false;

        return true;
    }
}
```